

CONTACT INFORMATION

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ABOUT

I am an ambitious and curious female Cameroonian researcher with a background in electronics and a double Masters in Physics and Electrical Engineering. Currently pursuing a PhD at the University of Dschang, my research focuses on the "Optical detection and classification of water-borne disease agents". I aim to develop deep learning models for detecting and classifying microorganisms in water, food, and bodily fluids.

SPECIALITY AND FIELDS

Speciality: Theoretical and experimental Physics
Fields: Main: Optics, Image Processing, Computer Vision, Deep Learning, Artificial intelligence, Microcontroller programming.
Others: Electronics, Electrical engineering, Power Systems, Automatic Control Systems, Nanomaterials.

RESEARCH INTERESTS

Applied electronics and biological physics, Applied optics and optoelectronics, Image processing and deep learning for biomedical imaging, Microcontroller-based systems for microbial detection, Printed circuit board design and fabrication, Electrical and electronic systems installation and maintenance, Electrotechnics and automation for household and industrial protection, Photoluminescence characterisation of nanomaterials

TEACHING EXPERIENCE

- **African Institute of Mathematical Sciences (AIMS), Cameroon**, Masters program, Teaching Assistant (Data Science and Climate Science streams) - 2025/2026
- **University of Dschang**, Master I Metrology and Biomedical Engineering, Biomedical Electronics, Second semester - 2024/2025
- **University of Dschang**, Master I GEII, Optimization of electric energy systems, First semester – 2023/2024
- **ISSTN Nanfah, Dschang**, Master I GEII (Electrical networks, Semiconductor Physics and high-frequency circuits), First semester - 2023/2024
- **Government Technical Bilingual High School Mbandjock**, Electrical Power System Department – 2022 till date.

CURRENT ACADEMIC POSITION

- **Ph.D. student in Physics**, Faculty of Sciences, University of Dschang, since November 2021 to present.
- Supervisors: **Pr. TAMO Tatietse Thomas**, Professor, University of Yaounde I and **Pr. TALLA MBE Jimmi Hervé**, Associate Professor, University of Dschang.

EDUCATION

University of Dschang, Cameroon

M.Sc., Physics, *Speciality: Electronics, Electrical Engineering and Automation*, 2021.

- Establishment: *Faculty of Sciences, University of Dschang, Cameroon.*
- Thesis Topic: *Optical Detection of Water-borne disease agents : Case of E-coli and Total coliform.*
- Supervisor: **Pr. TALLA MBE Jimmi Hervé**, Associate Professor, University of Dschang.
- Area of Study: Optics, Electronics, Electrical Engineering, and Automation.

Technical School Teacher's Diploma (DIPET II), Electrical and Power Engineering, *Speciality: Electrotechnics*, 2021.

- Establishment: *Higher Technical Teacher Training College (HTTTC), University of Bamenda.*
- Thesis Topic: *Design and Implementation of Over Voltage/Under voltage Protection System For Single Phase Networks.*
- Supervisor: **Pr. Asoh Derek Ajesam and Mr. Tsokezo T.**, University of Bamenda.
- Area of Study: Electrical protection systems.

Bachelors degree in Technology, Electrical Engineering, 2019.

- Establishment: *University Institute of Technology (IUT) FOTSO Victor Bandjoun, University of Dschang.*
- Topic: *Iris Recognition System.*
- Supervisor: **Pr. Tchiotsop Daniel, Professor**, University of Dschang.
- Area of Study: Biometrics, Image processing, Electronics and Electrotechnics.

University Diploma of Technology, Electrical Engineering, 2018.

- Establishment: *University Institute of Technology (IUT) FOTSO Victor Bandjoun, University of Dschang.*
- Area of Study: Electronics.

CERTIFICATIONS

1. Open and Distance Education of the Federal University of Viçosa, Brazil
Course on Image Processing and Applications
September 19th 2025. **Total workload:** 45 hours.
2. Deep Learning specialization: **Convolutional Neural Networks.**
DeepLearning.AI-Coursera August 25th 2024.
3. Improving Deep Neural Networks: **Hyperparameter Tuning. Regularization and Optimization.**
DeepLearning.AI-Coursera May 21st 2024.
4. Deep Learning Specialization: **Structuring Machine Learning Projects.**
DeepLearning.AI-Coursera May 21st 2024.

5. Deep Learning Specialization: **Neural Networks and Deep Learning.**
DeepLearning.AI-Coursera November 22nd 2023.
6. Machine Learning Specialization: **Supervised Machine learning - Regression and Classification.**
DeepLearning.AI-Coursera October 29th 2023.

REFEREED
JOURNAL
PUBLICATIONS

1. Design and Implementation of an Automatic Over/Undervoltage Protection System for Single Phase Low Voltage Power Lines.
Journal of Power and Energy Engineering (2022).
DOI: 10.4236/jpee.2022.108002.
2. Direct detection and classification of E. coli and fecal streptococci using optical water droplet method and convolutional neural networks.
Applied Optics.
DOI: 10.1364/AO.542072.
3. Classification of bacterial images obtained optically using some pre-trained models.
Journal of Microbiological Methods (Elsevier)
DOI: <https://doi.org/10.1016/j.mimet.2026.107572>

CONFERENCES,
SEMINARS,
WORKSHOPS

1. 4th African Conference on Fundamental and Applied Physics, **ACP-2025**
Poster Presentation Topic: Direct detection and classification of E. coli and fecal streptococci using an optical water droplet method and convolutional neural networks
University of Lomé, Togo September 14th to 20th 2025
2. Oral presentation for the 3th edition symposium **Women in STEM, and innovation for a sustainable future**
Organized by the: **Faculty of Sciences, University of Dschang, Cameroon**
March 27th 2025
3. 6th International Conference on Soft Materials **ICSM-2024**
Organized by the: **Rovira i Virgili University and the Soft Materials Research Society (SMRS)**
Universitat Rovira i Virgili, Tarragona, Spain November 17th to 20th 2024
4. Conference organized by the **Cameroon Physical Society**
Theme: **"High Level Physics and appropriate solutions to real life problems in developing countries"**
(**Poster Presentation Topic:** Characterization and modeling of water-borne disease agents interacting with light)
University of Dschang, Cameroon December 5th to 8th 2023

5. Conference organized by the **African Optical Fiber Family**
Theme: "**L'ingenierie de telecommunication et les nouvelles techniques de construction des prototypes d'ondes et signaux susceptibles de propager dans les guides d'ondes: techniques analytiques de construction et de numerisation des signaux**"
University of Dschang, Cameroon September 1st 2023
6. Workshop on Made in Cameroon for Industrialization and Sustainable Economic Development.
Cameroon Academy of Young Scientist May 12th 2022.
7. Finalist on Three Minutes to Convince CAYS .
Cameroon Academy of Young Scientist. May 27th 2022.
8. Internship on Computer Hardware maintenance, Installation and maintenance of a Smart-lock System.
Informatique et Génie Electrique Camerounais. August 2020.
9. Internship on installation and maintenance of Electronic and Electrical Appliances.
Générale des compagnies industrielles (GCI). May - June 2018.

PROJECTS

- *Wavelength Division Multiplexing*
- *Synthesis of a Buck-Boost converter*
- *Intelligent energy meter billing system based on a GSM Module*
- *Study of a Hopfield dynamic system: Numerical analysis and experimental verification*
- *Design and implementation of a drone under IEEE Cameroon student chapter and sponsored by IEEE photonics society*

HONOURS AND AWARDS

- **Best oral presentation award** for the 3rd Symposium for **Women in STEM** - Faculty of Sciences, University of Dschang, Cameroon - 27 March 2025
- **Coimbra group scholarship for young African Scientists**, University of Barcelona, Spain - November 2024 - January 2025
- Regional scholarship "**Women and Sciences**"
Congolese foundation for medical research - 2023/2024
- Award of excellence by the president of the Republic of Cameroon
University of Dschang - 2018/2019 – 2021/2022
- Award of excellence by the president of the Republic of Cameroon
University of Bamenda - 2019/2020– 2020/2021

COMPUTING:

- C++, **Python**, MATLAB, **Arduino**, MicroC for PIC, PSpice, **Proteus**, Solidworks.
- Desktop Editing and Productivity Software:**
 - $\text{\LaTeX}$ ($\text{\LaTeX}$, $\text{\BIBTeX}$)
- Operating Systems:**
 - Windows, Linux, and other UNIX variants.

REFERENCES
AVAILABLE TO
CONTACT

Microsoft:

- Word, Excel and PowerPoint.

Pr. TALLA MBE Jimmi Hervé (email)

- Associate Professor,
Department of Physics, Faculty of Science, University of Dschang, P.O. Box 96, Cameroon,
- Masters supervisor and PhD co-supervisor

Pr. TAMO Thomas Tatietse (e-mail: thomas2t@yahoo.fr)

- Professor,
National Advanced School of Engineering of Yaounde, University of Yaounde I, P.O. Box 8390, Cameroon,
- PhD supervisor and former Director of the University Institute of Technology Fotso-Victor (IUT-FV)

Pr. Gemma Piella (email)

- Professor,
Department of Engineering, Universitat Pompeu Fabra, Barcelona, Spain,
- Research collaborator

Pr. ASOH Derek (email)

- Associate Professor,
Higher Technical Teacher Training College (HTTTC), University of Bamenda, P.O. Box 39 Bambili, NW Region - Cameroon,
- Department of Electrical/Electronic Engineering, University of Bamenda, P.O. Box 39 Bambili, NW Region - Cameroon.
- Research collaborator

Pr. Frank Güell Vilà, (email)

- Professor,
Department of Electronics and Biomedical Engineering, Faculty of Physics, University of Barcelona, Martí i Franquès, 1 08028 Barcelona,
- Supervisor for the Coimbra Group scholarship programme.

Pr. Kenfack Sadem Christian, (email)

- Associate Professor,
Unité de Recherche de Matière Condensée, d'Electronique et de traitement du Signal (URMACETS), Dschang School of Science and Technology, University of Dschang, P.O. Box 96, Cameroon,
- Head of URMACETS.

LANGUAGE
SKILLS
HOBBIES

Fluent in speaking, writing and reading **English** and **French**.

Music: singing and dancing, tourism, reading, sports.

Dschang, August 10, 2026

CHIA Lizette Nange